QP CODE: 24900177

MAHATMA GANDHI UNIVERSITY

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First semester

Discipline Specific Core Course-MG1DSCMBG100 –UNSEEN WORLD OF MICROBES

Maximum Marks: 50

1. **State spontaneous generation theory [K][1]**

Definition of the Theory (1 Mark), Disproof of the Theory (1 Mark)

Mention of at least one key experiment or scientist (e.g., Pasteur, Redi, or Spallanzani) who disproved the theory.

2. **List out the contributions of Antony van Leeuwenhoek [K][1]**

Mention at least two key contributions (1+1)

3**. Name the theory disproved by Louis Pasteur? [K][1]**

Spontaneous Generation Theory- 1mark

Explanation- 1 mark

4. **Give example of a vaccine and mention its use [K][1]**

Example-1 mark, use 1 mark

5**. How does the domain Bacteria differ from Archaea? [K][2]**

Mentions at least two major differences - cell wall composition (1) membrane lipids (1)

6. Define Prokaryotes [U][3]

Definition - absence of a true nucleus and membrane-bound organelles. (1+1)

7. Explain the Functions of flagellum [U][3]

Locomotion (1) and sensory functions (1)

8. Define Mesosomes? [U][3]

Definition (1), Any one function (1)

9 Explain the functions of Ribosomes[U][3]

Any 2 functions (1+1)

10. Write a note on Bacterial cell wall [U][3]

Presence of peptidoglycan (1) one function (1)

11. Give notes on Mitochondria [U][3]

Structure-(1) , Function (1)

Part B

Answer any 4 out of 6, 5 marks each

12. List out Koch's Postulates [K][1]

Four postulates with descriptions (4) , Mention the experiment (1)

13**. Describe the Discovery of Penicillin [K][1]**

The year of discovery (1928) , Alexander Fleming (1)

Explain -The accidental nature of the discovery and the identification of penicillin. (4)

14**. As a microbiology student suggest method to improve crop yield using microbes [K][1]**

Answer any 5 microbial uses to enhance crop yield.

Eg. Biofertilizers:- Rhizobium, Azotobacter, and Azospirillum , Mycorrhizal Fungi,

Biopesticides: Plant Growth-Promoting Rhizobacteria (PGPR), Compost and Organic Matter Degradation: Trichoderma and Aspergillus.

15. What are the features of three kingdom classification [K][2]

Basis of Classification (1) Names of three kingdom (1), Explanation (3)

16. Discuss in detail about the morphological characteristics of bacteria? [U][3]

Explain- Shape- Cocci , Bacilli ,Spirilla, Vibrio, Filamentous with example (1)

Arrangement- Arrangement -Diplococci , Streptococci , Staphylococci, Palisades: Bacilli arranged side-by-side (1), Flagella (1), capsule (1), Fimbriae, pili (1)

17. Briefly explain any 5 beneficial and harmful effects of microbes and provide examples for each.

[U][3]

5 Beneficial Effects (with examples) (2.5), 5 Harmful Effects (with examples) (2.5)

Part C

18. Explain in detail about recent trends in microbiology [K][2]

Introduction (1.5), Main part- Molecular Medicine (1.5), Personalized medicine(1.5) , Preventive medicine(1.5) , gene therapy(1.5) , antimicrobial peptides of microbial origin(1.5) , conclusion(1)

19. What are the main characteristics and features of viruses? [U][3]

Explain- Non-Cellular Nature, Obligate Intracellular Parasites, Host Specificity, Simple Structure, shape- : helical, icosahedral, or complex (bacteriophages).

Genetic material: DNA or RNA (single-stranded or double-stranded).

Protein coat (capsid), outer lipid envelope.

Examples: Bacteriophage, plant virus, Animal virus (8), relevant diagrams (2)